

AN ACTIVE AUSTRALIAN SCIENCE 10 CURRICULUM MODULE

AUS BS10 SU STRAND: MODULE 1: DNA AND GENETICS

STUDENT'S NAME:

TUTORIAL:

TEACHER'S NAME:

DATE:

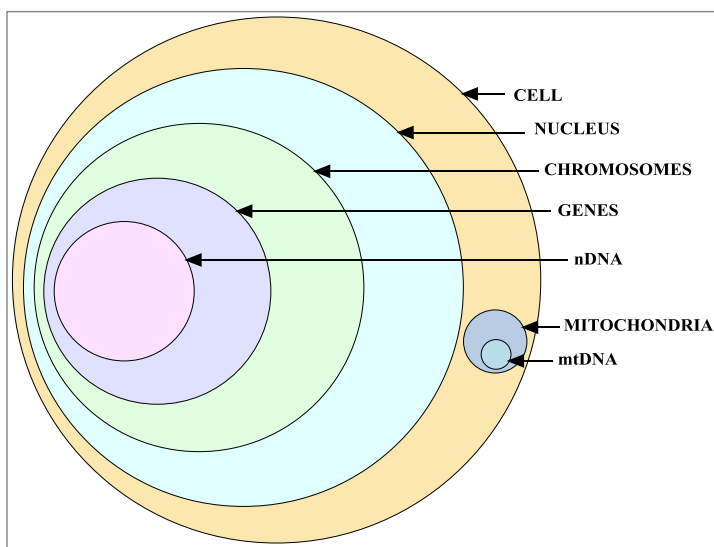
AUS Biological Sciences 10: Science Understanding Strand Module 1: Key Conceptual Understandings

- Students can, using a systems diagram, describe the relationship between each of these terms: cell, nucleus, chromosome, and gene, nuclear and mitochondrial DNA.
- Students know that DNA contains the genetic code that controls the production of proteins in living things. The genetic code (ATGC) is the set of instructions in a gene that tell the cell how to make a specific protein. Each human being, other than monozygotic individuals (identical twins, triplets etc.), has a unique genome and thus a personal DNA profile. The genome is the entire DNA contained in an organism or in a cell, which includes both the chromosomes within the cell nucleus and the DNA in the mitochondria and chloroplasts.
- Students understand that living organisms reproduce in a variety of ways that can be classified into two main types: asexual reproduction and sexual reproduction.
- Students understand that **sexual reproduction** is the production of offspring by combining genetic material from two parent organisms, which occurs when a male gamete **fertilises** a female gamete.
- Students understand that in sexual reproduction offspring are produced by the fusion of two cells called gametes.
- Students define **fertilisation** as the process in which two gametes fuse to form a zygote.
- Students know that genetics is the study of the patterns and mechanisms of inheritance and natural variation.
- Students know that inheritance is the passing on of inherited characteristics from parent(s) to offspring via the chromosomes they receive from their parent(s).
- Students understand the difference between genotype and phenotype. **Genotype** refers to the actual alleles an individual possesses (or genetic make-up) that determines a specific trait. **Phenotype** refers to the characteristics of that specific trait as displayed by the individual.
- Students understand that a **dominant allele** masks the effect of a recessive allele in the heterozygote.
- Students understand that the effect of a **recessive allele** is hidden by the effect of a dominant allele in the heterozygote.
- Students understand that an allele is a variety of a particular gene. For example, the gene for making red blood cell surface protein may contain instructions for making variety A protein, variety B protein, or no protein. Geneticists therefore say that this gene has 3 alleles, or 3 allelous forms.
- Students know that each of the alleles in an allelous pair need not be identical, with the alternative forms of the gene being referred to as **alleles**.
- Students understand that a **monohybrid cross** is a cross between two parents differing in one trait, e.g., blue or brown eyes.
- Students understand that a **Punnett square** is a diagram that geneticists use to show/predict all possible allele combinations in a cross (mating) of parents whose alleles are known.
- Students know that a **pedigree** is a chart of a person's (or other animal's) family tree showing the inheritance of a particular trait.

- Students understand that **pedigree analysis** is the determination of the pattern of inheritance of a trait, condition or disease by reference to a family tree in which the presence or absence of the particular trait is recorded over successive generations.
- Students know that **mutations** are sudden permanent changes in genes producing new alleles and genes.
- Students know that **mutants** are individuals that show the effect of a mutation in their phenotype.

From Cells to Genes

- **Backward Reference:** In the AUS Biological Sciences 8 SU Strand Module 1, you learnt that the nucleus of eukaryotic cells contains nuclear DNA ([nDNA](#)), and that the mitochondria in eukaryotic cells contain mitochondrial DNA ([mtDNA](#)).
1. Can you now describe the relationship between each of these terms: cell, nucleus, chromosomes, genes, nDNA, mitochondria, and mtDNA?

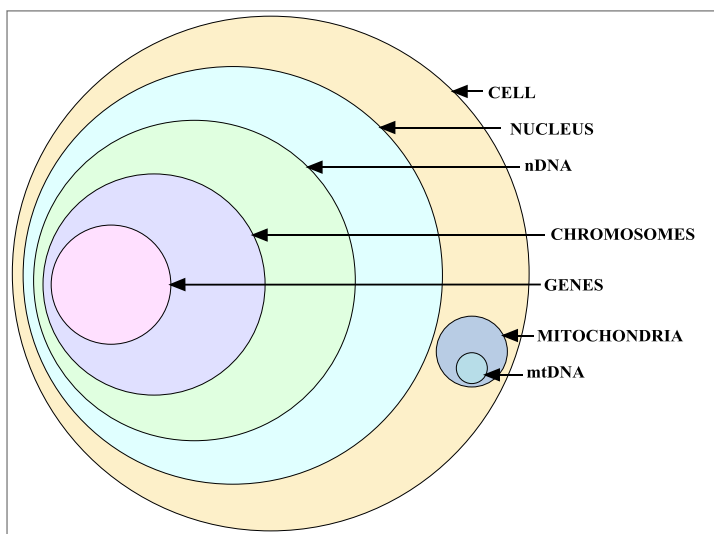


Systems Diagram 1

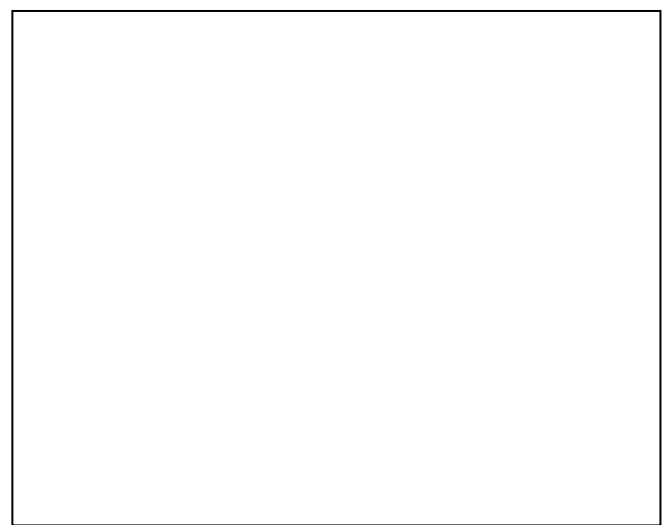


[6 marks]

2. Why is Systems Diagram 1 more correct than Systems Diagram 2? In what way is Diagram 2 incorrect?



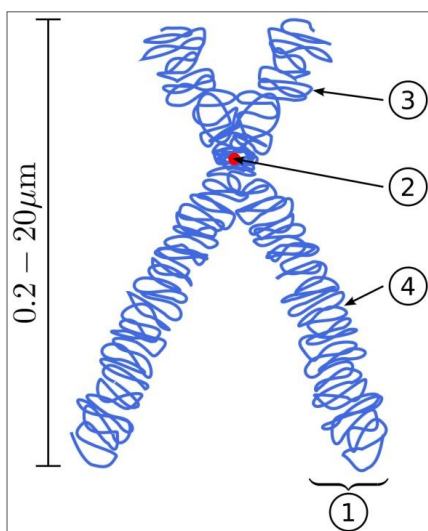
Systems Diagram 2



[3 marks]

- Science Concept: Cells are tiny living units that, together with cell products, make up an organism (or living thing).
- Science Concept: The cell nucleus contains the nucleolus, which is composed of proteins, DNA and RNA, and which makes ribosomes. Ribosomes read the genetic code of genes and use that code to assemble the appropriate protein.
- Science Concept: Chromosomes are big thread-shaped structures that contain nucleic acid (usually DNA), and, except in bacteria, a protein scaffold.
- Science Concept: A Gene is a linear code sequence of nucleotides A, T, G and C, along a segment of DNA that provides the instruction for synthesis of RNA, which, when translated into protein, leads to the expression of a hereditary trait.
- Science Concept: Nuclear DNA, nuclear deoxyribonucleic acid (nDNA), is the DNA contained in the nucleus of eukaryotic cells.
- Science Concept: Mitochondria are membrane-enclosed organelles found in most eukaryotic cells that convert energy from digested food, and oxygen, into chemical potential energy stored in adenosine triphosphate (ATP) molecules.
- Science Concept: Mitochondrial DNA (mtDNA) is the DNA contained within the mitochondria found in most eukaryotic cells. Chloroplasts, in green plants, similarly contain their own DNA.

Structure of a replicated Chromosome



Key to the adjacent sketch diagram of a replicated Chromosome:

- (1) Chromatid: A single strand of DNA after chromosome replication.
- (2) Centromere is part of the chromosome where two duplicated copies of DNA are still attached, prior to the final stages of mitosis or meiosis, where they separate into separate daughter cells.
- (3) Short arm = p arm, and (4) Long arm = q arm.

$1.0 \mu\text{m} = 1.0 \text{ micrometre} = 1.0 \times 10^{-6} \text{ metres.}$

Source of Image URL: <http://en.wikipedia.org/wiki/Chromosome>

3. Under what conditions are chromosomes of a eukaryotic cell visible in a light microscope?

[3 marks]

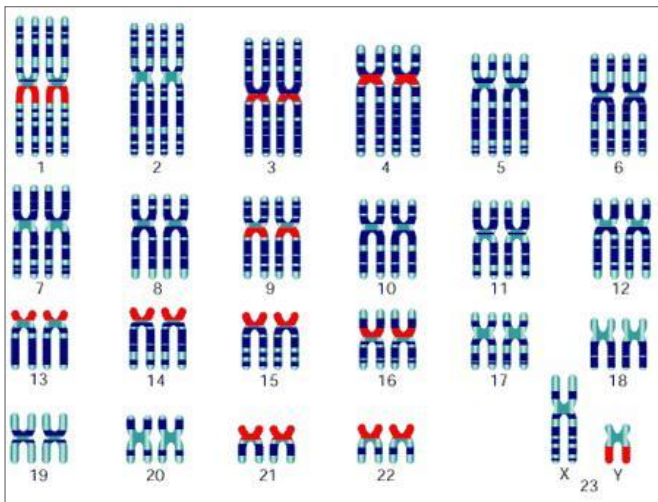
Chromosomes are not visible within a cell's nucleus, not even under a light-microscope, when the cell is not dividing. However, the DNA that makes up chromosomes becomes more tightly packed during cell division, and is then visible under a light-microscope. Science websites and textbooks that show cell nuclei containing chromosomes, should state that the cells, depicted in displayed diagrams, are undergoing cell division.

The Human Karyotype

- Science Concept: A [karyotype](#) is the number and appearance under a light microscope of chromosomes in the nucleus of a [eukaryotic cell](#). The term is also used for the complete set of chromosomes in a species, or an individual organism, e.g., human body cells normally contain 23 pairs of chromosomes, i.e., 46 chromosomes in each cell.

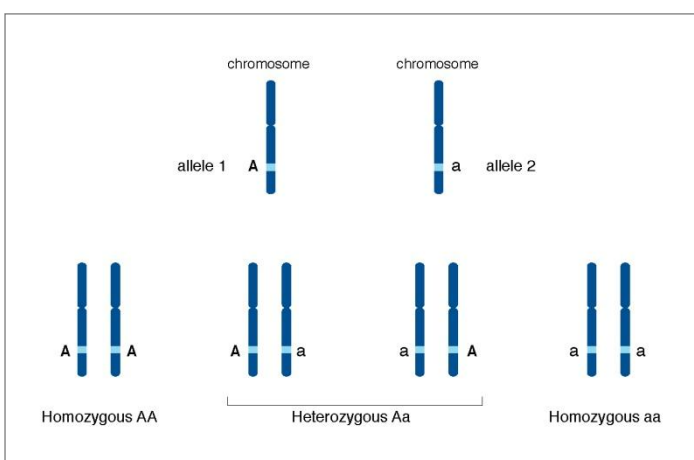
Human females have 22 pairs of [autosomes](#) (non-sex chromosomes), and 1 pair of [sex chromosomes](#) (XX). Human males have the same 22 pairs of autosomes (non-sex) chromosomes, but a different 'pair' of sex chromosomes ([X](#) and [Y](#)).

- Is the karyotype (a diagrammatic sketch) shown below, possibly from a human? Is it from a male, or a female? Explain.



[4 marks]

Genes, Alleles, and Homologous Chromosomes



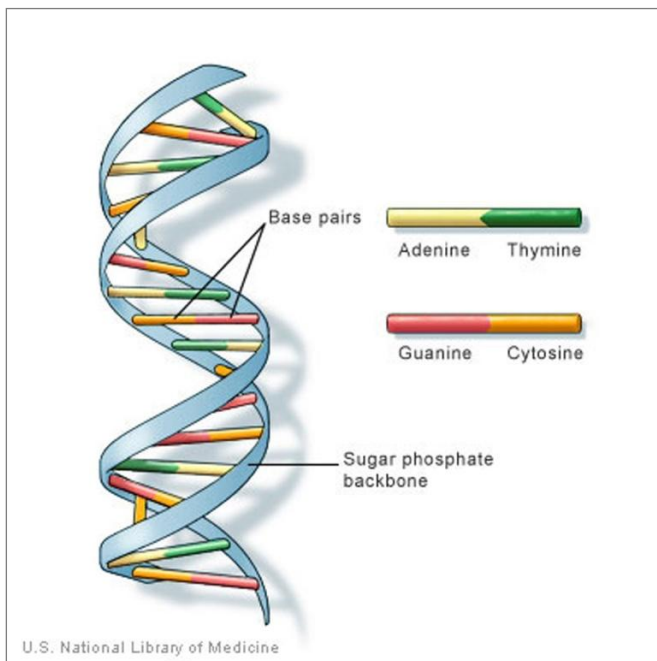
Science Concept: Humans and many other species can have no more than two alleles for a given gene, each allele residing on a separate, homologous chromosome.

Science Concept: An [allele](#) is one of two or more forms of a gene situated in the same position (called a locus) on homologous chromosomes.

Source of Image URL: <http://www.genome.gov/glossary/?id=4>

Structure of DNA

5. Can you now describe the structure of DNA? What are [base pairs](#)? What is a [sugar phosphate backbone](#)?



[4 marks]

Source of Image URL: <http://ghr.nlm.nih.gov/handbook/basics/dna>

- Science Concept: The [genome](#) is the entire DNA sequence contained in an organism or in a cell, which includes both the chromosomes within the cell nucleus and the DNA in mitochondria and chloroplasts.

Sexual Reproduction

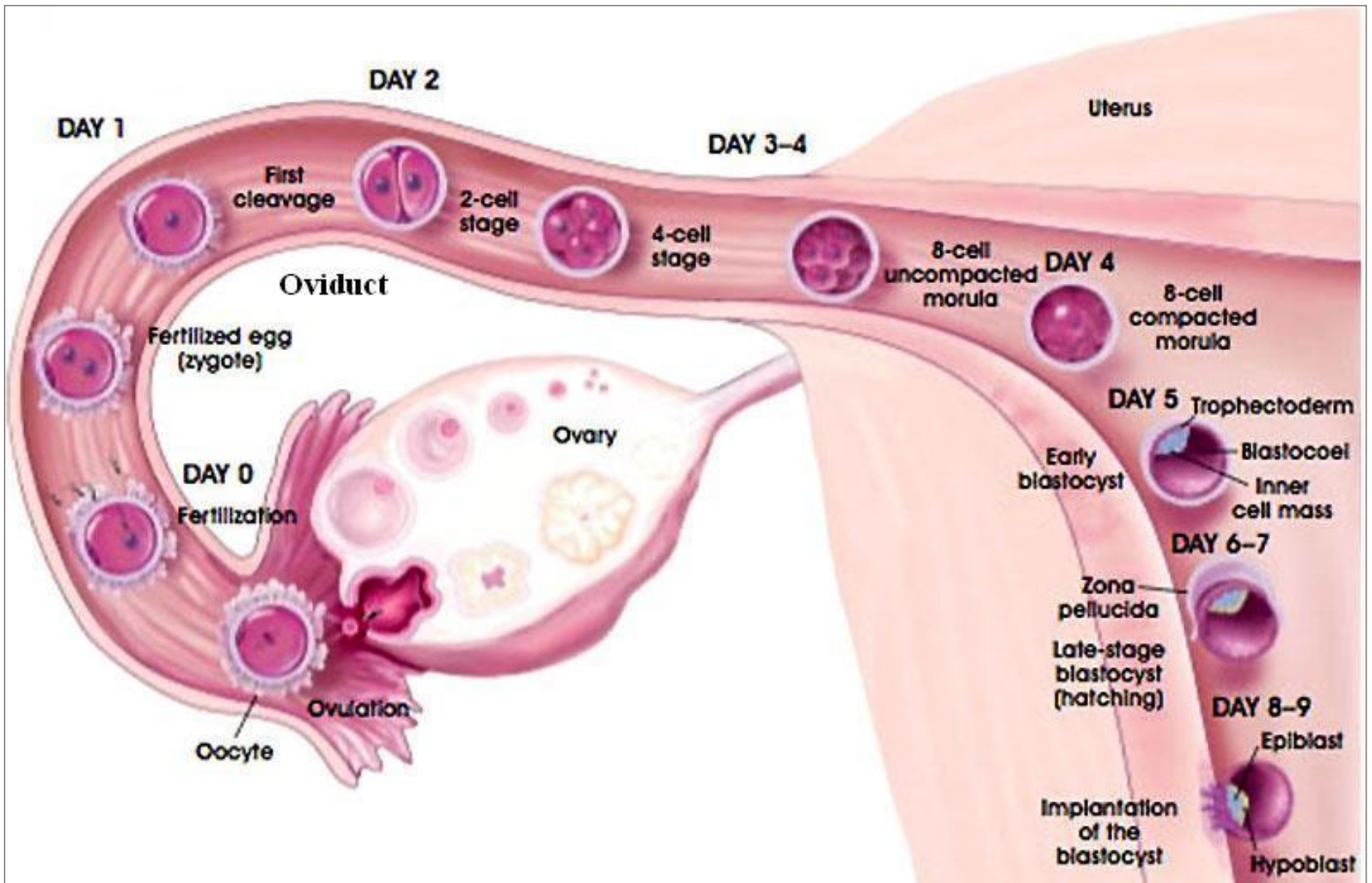
- Science Concept: All organisms under suitable conditions reproduce more individuals of their own kind to ensure survival of the species. **Reproduction** is the process by which a species ensures its survival.
- Science Concept: Living organisms reproduce in a variety of ways that can be classified into two main types: **asexual** reproduction and **sexual** reproduction. In **sexual reproduction** offspring are produced by the fusion of two sex cells called **gametes**.
- Science Concept: **Fertilisation** is the process in which two gametes fuse to form a **zygote**. Females produce egg cells or ova (female gametes) in their ovaries, whereas males produce sperm cells (male gametes) in their testes.

Fertilization

Ovum + Sperm → Zygote (Fertilized ovum) → Morula → Blastocyst → Embryo

- Science Concepts: The [blastocyst](#) is a structure formed in the [early embryonic development](#) of mammals, after the formation of the morula. A [morula](#) is an embryo, at an early stage of embryonic development, which consists of cells (called blastomeres) in a solid ball, which is contained within the zona pellucida.
- Science Concept: An [embryo](#) is the early developing stage of an organism formed as a result of division of a zygote.

The Normal Site of Human Fertilization



Source of Image URL: http://php.med.unsw.edu.au/embryology/index.php?title=Main_Page

Now study the diagram, given directly above, to locate where the sperm normally fertilizes the ovum.

6. In precisely what part of the female reproductive system does a sperm normally fertilize an ovum?

[2 marks]

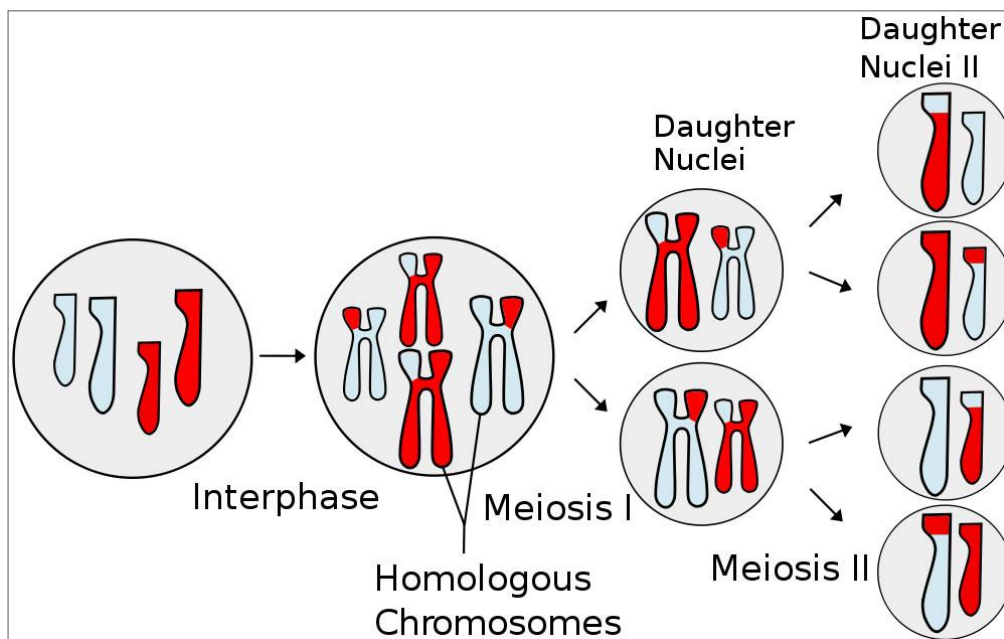
- Science Concepts: An **ovary** is the organ that produces female gametes (ova) in animals, and (ovules) in plants. **Ovulation** is the release of an ovum from the ovary. A **zygote** is a fertilized ovum (or egg).

7. What process (behavioural adaptation) do male and female humans normally use to fertilize an ovum?

[2 marks]

A Schematic Diagram of Meiosis

- Science Concept: [Meiosis](#) is a two-part cell division process of those organisms that sexually reproduce. Meiosis results in sex cells (or gametes) that contain half the number of chromosomes of the parent cell, as shown in the following schematic diagram.



Source of Image URL: <http://en.wikipedia.org/wiki/Meiosis> [This diagram shows [chromosomal crossover](#).]

In plants, spores are made by meiosis, whereas in animals, gametes are made. Meiosis involves two separate cell divisions shown as Meiosis 1, and Meiosis II, in the schematic diagram. The first cell division separates the [homologous pairs of chromosomes](#) (each consisting of two chromatids) into different cells. Each of these new cells then undergoes a second cell division, where the chromatids are separated into four daughter cells, each with half the chromosome number of the parent cell.

- Science Concept: A [chromatid](#) is one of the two identical copies of DNA, joined at their [centromeres](#), which make up a duplicated chromosome. Please refer back to the chromosome sketch, given on Page 3. Before cell division when the [DNA code](#) is being read, chromosomes consist of a single strand of DNA. At the very start of cell division, the DNA makes an exact copy of itself to become two chromatids that remain joined at their [centromeres](#). Chromatids are called [sister chromatids](#) only while they remain joined at their centromeres. When chromatids separate during the final stages of mitosis or meiosis, the two strands of DNA are then referred to as [daughter chromosomes](#).
8. How are sister chromatids related to daughter chromosomes? You must explain your answer using brief typed notes and a labelled diagram.

[3 marks]

A Glossary of Genetic Terms

- Science Concept: Genetics is the study of patterns and mechanisms of inheritance and natural variation.
- Science Concept: Inheritance is the passing on of inherited characteristics from parent(s) to offspring via the chromosomes they receive from their parent(s).
- Science Concepts: **Genotype** refers to the actual alleles an individual possesses (or genetic make-up) that determines a specific trait. **Phenotype** refers to the characteristics of that specific trait as displayed by the individual.
- Science Concept: A **dominant allele**, e.g., (A) masks the effect of a recessive allele in the heterozygote.
- Science Concept: The effect of a **recessive allele** (a) is hidden by the effect of a **dominant allele** (A) in the heterozygote.
- Science Concept: An [allele](#) is a variety of a particular gene. For example, the gene for making red blood cell surface protein may contain instructions for making variety A protein, variety B protein, or no protein. Geneticists therefore say that this gene has 3 alleles, or 3 allelous forms. An [allele](#) is one of two or more forms of a gene situated in the same position (called a locus) on homologous chromosomes.
- Science Concept: A monohybrid cross is a cross between two parents differing in one trait, e.g., blue or brown eyes.
- Science Concept: A Punnett square is a diagram that geneticists use to show/predict all possible allele combinations in a cross (mating) of parents whose alleles are known.
- Science Concept: A pedigree is a chart of a person's (or other animal's) family tree showing the inheritance of a particular trait.
- Science Concept: Pedigree analysis is the determination of the pattern of inheritance of a trait, condition or disease by reference to a family tree in which the presence or absence of the particular trait is recorded over successive generations.
- Science Concepts: Mutations are sudden permanent changes in genes producing new alleles and genes. Mutants are individuals that show the effect of a mutation in their phenotype.

Mechanisms of Inheritance

An Austrian monk, [Gregor Johann Mendel](#) (1822-1884), showed that an inherited characteristic is determined by the combination of a pair of hereditary units, or genes, one from each of the parental sex cells, or gametes. [Mendel's Laws](#), stated in modern terms are:

- The [law of segregation](#) states that when meiosis occurs the members of each original gene pair separate into separate gametes.
- The [law of independent assortment](#) states that pairs of genes separate independently of each other during gamete formation. However, geneticists now know that the law of independent assortment is only strictly true for gene pairs located on different chromosomes.

In 1865, Mendel presented the results of his breeding experiments with the garden pea, *Pisum sativum*, to the Brunn Natural Science Society. His experimental data and conclusions were published by the Society in 1866, but his work was ignored until 1900, when three other scientists independently obtained similar data.

9. Why was Mendel's research with the breeding of pea plants ignored until sixteen years after his death?

[2 marks]

Dominant and Recessive Inheritance in Humans

TABLE 1: Dominant and Recessive Inheritance in Humans (can be different when rare alleles are involved)

Inherited Characteristic or Trait	Dominant	Recessive
Type of earlobes	Free	Fixed
Ability to roll tongue into U shape	Can roll tongue	Can't roll tongue
Type of hairline	Has Widow's Peak	Doesn't have Widow's Peak
Relative length of second toe	Longer than big toe	Shorter than big toe
Relative position of thumbs	Left thumb on right thumb	Right thumb on left thumb
Eye colour	Dark eyes	Blue eyes
Skin pigmentation	Normal	Albinism
Length of fully grown limbs	Achondroplasia (Dwarfism)	Normal length of limbs
Number of digits	Polydactylism (extra digits)	Normal number
Visual acuity	Near or Farsightedness	Normal vision
Red/Green colour vision	Normal colour vision	Colour blindness
ABO blood typing	A or B blood factor	O blood factor
Type of haemoglobin	Normal haemoglobin	Sickle haemoglobin

Tongue Rolling in Humans

The allele for tongue-rolling **R** is dominant over the allele for non tongue-rolling **r**, as shown in TABLE 2.

TABLE 2: Tongue Rolling in Humans

Genotype	Phenotype
(R R) Homozygous dominant (Pure breed)	Tongue Roller
(R r) Heterozygous (Hybrid)	Tongue Roller
(r r) Homozygous recessive (Pure breed)	Non Roller

10. If you can roll your tongue, are you a homozygous dominant (pure breed) tongue roller (**R R**), or a heterozygous (hybrid) tongue roller (**R r**)? Can you now show, by completing the two labelled Punnett squares, how you could determine whether you are a pure or hybrid tongue roller? HINT: To answer this question, you will need to assume that you have children with a non tongue-rolling partner. You must also explain your answer using brief notes typed into the textbox, provided beneath the Punnett squares.

MALE PARENT GENOTYPE

MALE PARENT PHENOTYPE

MALE GAMETES

FEMALE PARENT PHENOTYPE

FEMALE PARENT GENOTYPE

FEMALE GAMETES

OFFSPRING GENOTYPE & PHENOTYPE

OFFSPRING GENOTYPE & PHENOTYPE

[20 marks]

Inheritance of Sex in Humans

- Science Concepts: [Sexual reproduction](#) is the production of offspring by combining genetic material from two parent organisms, which occurs when a male gamete fertilises a female gamete. [Fertilisation](#) is the process in which a male gamete fuses with a female gamete to form a zygote. The sex of a child is genetically determined at **conception**, i.e., when fertilisation occurs.

Human females have 22 pairs of [autosomes](#) (non-sex chromosomes), and 1 pair of [sex chromosomes](#) (X X). Human males have the same 22 pairs of autosomes (non-sex) chromosomes, and 1 different 'pair' of sex chromosomes (X Y).

11. What process is occurring at **C**? What is the genotype and phenotype of Zygote 1?

OVUM

22 + X

C

SPERM

22 + X

→

1

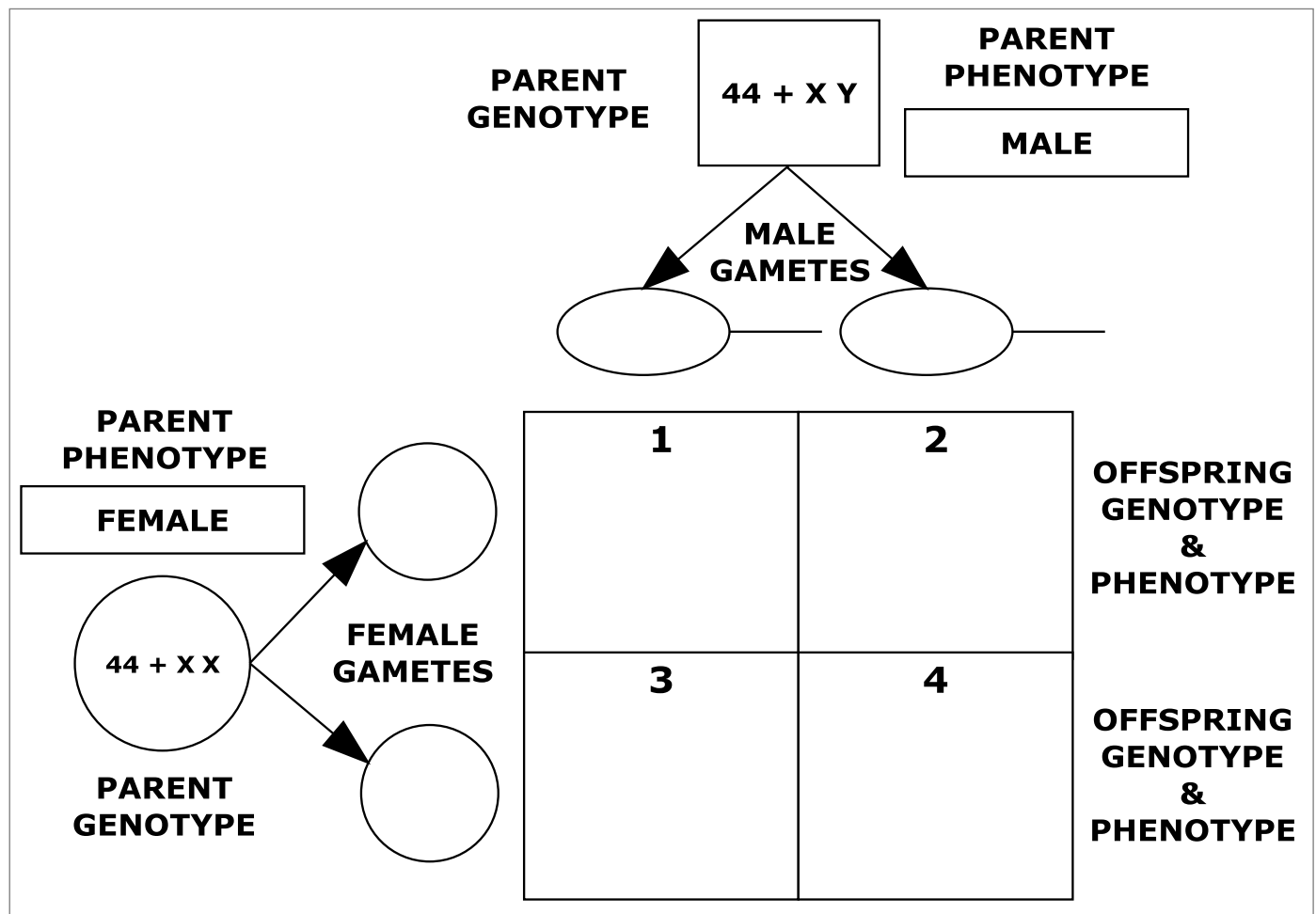
Genotype ?

Phenotype ?

[3 marks]

- Science Concept: [Meiosis](#) is a two-part cell division process of those organisms that sexually reproduce. Meiosis results in sex cells (or gametes) that contain half the number of chromosomes of the parent cell.
- Science Concept: Genotype refers to the alleles an individual possesses that determine a specific trait.

12. Can you now show, using the Punnett square, the probable genotypes and phenotypes of all offspring?



[6 marks]

13. Based on genetics only, what is the probability of conceiving a boy?

[1 mark]

14. Based on genetics only, what is the probability of conceiving a girl?

[1 mark]

X-Linked Recessive Inheritance in Humans

- Science Concept: [Mutations](#) are sudden permanent changes in genes producing new alleles and genes.
- Science Concept: [X-linked recessive inheritance](#) is a mode of inheritance in which a mutation in a gene on the X chromosome causes expression of the mutation in the phenotype. Males will always express such a mutation, since they have only one X chromosome (they are [hemizygous](#)). Females will only express a recessive mutation if they are homozygous for that mutation, i.e., they have the recessive allele on both X chromosomes. Red-Green colour blindness, haemophilia, and muscular dystrophy are classic examples of X-linked recessive conditions in humans.

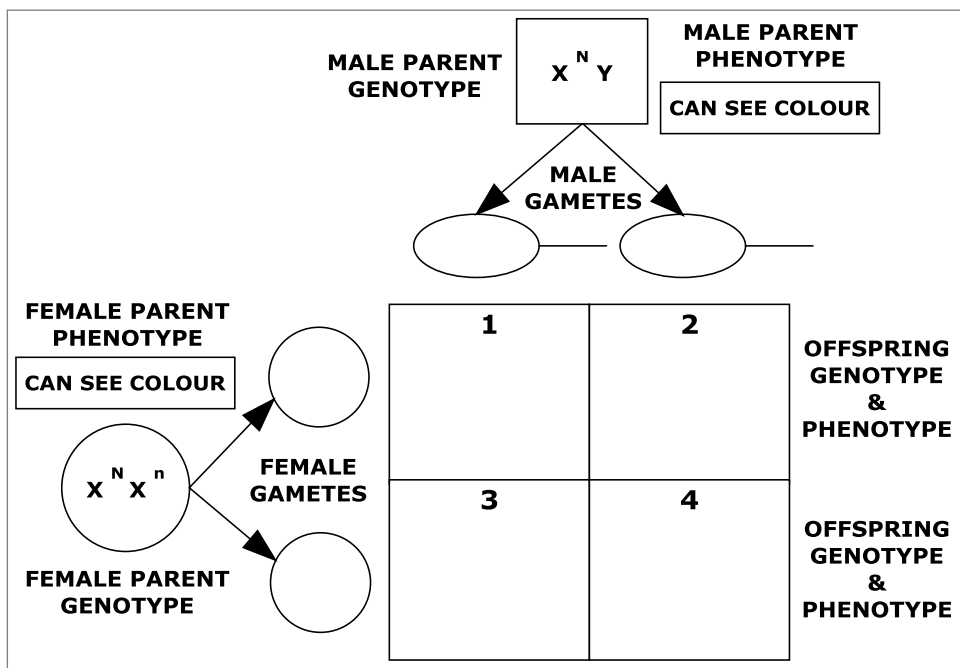
Red-Green Colour Blindness in Humans

Hereditary [red-green colour vision defects](#) show up in early childhood, mostly in males. Among Caucasians, about 8% of males and 0.5% of females have red-green colour vision defects; these defects are less frequent among African males (3% to 4%), and Asian males (3%). Detection of [severe red-green colour blindness](#), at high school age, should be communicated to parents and to affected children because the detection of colour vision defects may be relevant for certain occupational choices.

TABLE 3: Red-Green Colour Blindness

Male Genotype	Male Phenotype	Female Genotype	Female Phenotype
$X^N Y$	Can see Colour	$X^N X^N$	Can see Colour
$X^n Y$	Colour Blind	$X^N X^n$	Can see Colour
		$X^n X^n$	Colour Blind
N = Allele for Normal Vision		n = Allele for Colour Blindness	

15. Why do about sixteen times more Caucasian males than females suffer from colour blindness? You must support, and explain your answer by completing the Punnett square, and by typing notes into the textbox.



[10 marks]

Pedigree Analysis in Humans

- Science Concept: [Pedigree analysis](#) is the determination of the pattern of inheritance of a trait, condition or disease by reference to a family tree in which the presence or absence of the particular trait is recorded over successive generations.
- Science Concept: A pedigree is a chart of a person's (or animal's) family tree showing inheritance of a particular trait.

Human Genetics

- Science Concept: [Human genetics](#) is mainly concerned with the determination of the function of all genes found within the [genome](#), which will result in identification and understanding of those genes that cause human disease.

16. Which [genetic disorder](#) (or disease) will you select, study and report on? You must write, and include a pedigree analysis question, on your selected genetic disorder (or disease), in your MS Word or PP report.

Your report must answer all the focus questions, dot-pointed in Table 4, for your selected genetic disorder.

TABLE 4: Select your Genetic Disorder or Disease

TABLE 4: Select your Genetic Disorder or Disease			
Condition	Click to Tick	Genetic Disorder or Disease	Focus Questions
1		Albinism	• What is it? • What causes it? • How is it inherited? • How is it diagnosed? • How is it treated? • What is it like to have it? • Which web sites did you use in your research?
2		Achondroplasia (Dwarfism)	
3		Polydactylism (extra digits)	
4		Red-Green Colour Blindness	
5		Sickle Cell Anaemia	
URL 1: http://en.wikipedia.org/wiki/Albinism URL 2: http://en.wikipedia.org/wiki/Achondroplasia URL 3: http://en.wikipedia.org/wiki/Polydactyly URL 4: http://en.wikipedia.org/wiki/Sickle-cell_disease			

[10 marks]

Extra Online Resources



You may like to access these extra URL's to learn more about DNA and Genetics?

- URL 1: <http://www.nature.com/scitable>
URL 2: <http://www.genetics.edu.au/factsheet>
URL 3: <http://learn.genetics.utah.edu/content/begin/traits/>
URL 4: http://anthro.palomar.edu/abnormal/abnormal_5.htm
URL 5: <http://www.biotechnologyonline.gov.au/human/genedisorders.html>
URL 6: http://www.biotechnologyonline.gov.au/popups/int_geneprobe.html
URL 7: http://www.ornl.gov/sci/techresources/Human_Genome/medicine/genetherapy.shtml
URL 8: http://highered.mcgraw-hill.com/sites/0072485949/student_view0/chapter3/interactive_activity.html



What new science ideas have you now learned about **Genetics** by studying this AUS BS10 SU Strand Module 1? List your newly learned key Science Concepts, as dot-points, in the textbox given below.

A Self-Check Summative Assessment Table

Now complete this assessment table by adding the 'self-check' mark achieved for your answer to each focus question. Your teacher will provide his/her 'model' answer to each question, via 'whole of class' discussion.

Total Mark Possible = **80**

Mark Achieved =

Percent (%) Score Achieved =